HW-2. 01257001 宋辰星.

1.

(a) $f(n) = \pm n$: Not a function ($f(2) = 2$ or -2)

(b) $f(n) = \sqrt{n^2+1}$: Yes.

(c) $f(n) = 1/(n^2-4)$: Not a function (对 $n = \pm 2 \Rightarrow$ 未定義)

2.

(a)

assuming that $g(a) = g(b)$

$$f(g(a)) = f(g(b))$$

$$f \circ g(a) = f \circ g(b)$$

Since $f \circ g$ is one-to-one, $a = b$. Thus, by defn, g is one-to-one.

(b)

For example:

$A = \{0\}$, $B = \{0, 1\}$, $C = \{0\}$, $f: B \to C$, $g: A \to B$.

and $\begin{cases} f(0) = 0 \\ f(1) = 0 \\ g(0) = 0 \end{cases}$, then because $\forall c \in C, \exists b \in B$ s.t. $f(b) = c$, f is onto.

and because $\forall c \in C, \exists a \in A$ s.t. $f \circ g(a) = c$, $f \circ g$ is onto.

However, because $1 \in B$ s.t. $f(x) \neq 1 \; \forall x \in A$, g is not onto.

3.

(a)

$a_0 = \underline{2}$

$a_1 = 6a_0 = \underline{12}$,

$a_2 = 6a_1 = \underline{72}$.

$a_3 = 6a_2 = \underline{432}$

$a_4 = 6a_3 = \underline{2592}$

$a_5 = 6a_4 = \underline{15552}$

(b)

$a_0 = \underline{1}$

$a_1 = \underline{1}$

$a_2 = 2a_1 + 4a_0 = \underline{6}$.

$a_3 = 3a_2 + 9a_1 = 18 + 9 = \underline{27}$

$a_4 = 4a_3 + 16a_2 = 108 + 96 = \underline{204}$

$a_5 = 5a_4 + 25a_3 = 1020 + 675 = \underline{1695}$.

(c)

$a_0 = \underline{1}$

$a_1 = \underline{2}$

$a_2 = \underline{0}$

$a_3 = a_2 + a_0 = \underline{1}$

$a_4 = a_3 + a_1 = \underline{3}$.

$a_5 = a_4 + a_2 = \underline{3}$.

(d)

$a_0 = -1$

$a_1 = 0$.

$a_2 = 2a_1 + (a_0)^2 = 1$

$a_3 = 3a_2 + (a_1)^2 = 3$.

$a_4 = 4a_3 + (a_2)^2 = 13$.

$a_5 = 5a_4 + (a_3)^2 = 65 + 9 = 74$.

(e)

$a_0 = 1$

$a_1 = 1$

$a_2 = 2$.

$a_3 = a_2 - a_1 + a_0 = 2$.

$a_4 = a_3 - a_2 + a_1 = 1$

$a_5 = a_4 - a_3 + a_2 = 0$.

4.

(a) $a_n = 2a_{n-1} - 3$, $a_0 = -1$

4

$a_n = 2a_{n-1} - 3$

$= 2(2a_{n-2} - 3) - 3 = 4a_{n-2} - 9 = 2^2 a_{n-2} - (3 + 3\times 2)$

$= 4(2a_{n-3} - 3) - 9 = 8a_{n-3} - 21 = 2^3 a_{n-3} - (3 + 3\times 2 + 3\times 4)$

$= 8(2a_{n-4} - 3) - 21 = 16a_{n-4} - 45 = 2^4 a_{n-4} - (3 + 3\times 2 + 3\times 4 + 3\times 8)$

$$\begin{array}{r} 24 \\ 12 \\ 6 \\ 3 \\ \hline 45 \end{array}$$

$$a_n = 2^n a_{n-n} - \sum_{i=0}^{n-1} 3\times 2^i$$

$$= 2^n a_0 - \sum_{i=0}^{n-1} 3\times 2^i$$

$$= 2^n(-1) - 3\sum_{i=0}^{n-1} 2^i$$

等比級数

$$= -2^n - 3\frac{1(2^n - 1)}{2-1}$$

$$= -2^n - 3(2^n - 1)$$

$$= -2^n - 3\times 2^n + 3$$

$$= -4\times 2^n + 3$$

4-(b)

$$a_n = a_{n-1} + 2n + 3, \quad a_0 = 4$$

$$a_n = \underline{a_{n-1}} + 2n + 3 \qquad 2n-2+3+2n+3$$

$$= (a_{n-2} + 2(n-1) + 3) + 2n + 3 = \underline{a_{n-2}} + 4n + (3\times 2 - 2)$$

$$= (a_{n-3} + 2(n-2) + 3 + 2(n-1) + 3) + 2n + 3 = \underline{a_{n-3}} + 6n + (3\times 3 - 6)$$

$$2n-4+3+2n-2+3+2n+3$$

$$= (a_{n-4} + 2(n-3) + 3) + 2(n-2) + 3 + 2(n-1) + 3 + 2n + 3$$

$$+2n-6+3+2n-4+3+2n-2+3+2n+3$$

$$= \underline{a_{n-4}} + 8n + (3\times 4 - 12)$$

$$= a_{n-n} + 2\cdot\sum_{i=1}^{n} i + 3n$$

$$= a_0 + 2\left(\frac{n(n+1)}{2}\right) + 3n$$

$$= 4 + n^2 + n + 3n$$

$$= 4 + n^2 + 4n$$

$$= n^2 + 4n + 4$$

4-(C)

$a_n = 2n\, a_{n-1}, \quad a_0 = 1.$

$$
\begin{aligned}
a_n &= 2n\, a_{n-1} \qquad = 2n\, a_{n-1} \\
&= 2n\,(2(n-1)a_{n-2}) = 2\times 2\times n(n-1)a_{n-2} = 2^2 n(n-1)a_{n-2} \\
&= 2n\,(2(n-1)\,2(n-2)a_{n-3}) = 2\times 2\times 2\times n(n-1)(n-2)a_{n-3} = 2^3 n(n-1)(n-2)a_{n-3} \\
&= 2^n\,(n(n-1)(n-2)(n-3)\dots 1)\,a_{n-n} \\
&= 2^n \cdot n! \cdot a_0 \\
&= 2^n \cdot n! \cdot 1 \\
&= 2^n \cdot n!
\end{aligned}
$$

5.-(a)

$$a_n = -n+2$$

prove:

$$a_n = a_{n-1} + 2a_{n-2} + 2n - 9$$

$$\begin{cases} a_{n-1} = -(n-1)+2 = -n+3 \\ a_{n-2} = -(n-2)+2 = -n+4 \end{cases}$$

$$\begin{aligned} a_{n-1} + 2a_{n-2} + 2n - 9 &= (-n+3) + 2(-n+4) + 2n - 9 \\ &= -n+3 \cancel{-2n} \cancel{+8} \cancel{+2n} \cancel{-9} \\ &= -n+2 \\ &= a_n \end{aligned}$$

5-(b)

$$a_n = 3(-1)^n + 2^n - n + 2$$

prove:

$$a_n = a_{n-1} + 2a_{n-2} + 2n - 9$$

$$\begin{cases} a_{n-1} = 3(-1)^{n-1} + 2^{n-1} - (n-1) + 2 \\ a_{n-2} = 3(-1)^{n-2} + 2^{n-2} - (n-2) + 2 \end{cases}$$

$-n+4$

$$3(-1)^{n-1} + 2^{n-1} - (n-1) + 2 + 2\left(3(-1)^{n-2} + 2^{n-2} - (n-2) + 2\right) + 2n - 9$$

$$= 3(-1)^{n-1} + 2^{n-1} \cancel{-n+3} \cancel{+} 6(-1)^{n-2} + 2 \cdot 2^{n-2} \cancel{-2n+8+2n-9}$$

$$= 3(-1)^{n-1} + 2^{n-1} + 6(-1)^{n-2} + 2 \cdot 2^{n-2} - n + 2$$ (續下頁)

5. (b) (續).

$$= 3(-1)(-1)^{n-2} + 2\cdot 2^{n-2} + 6(-1)^{n-2} + 2\cdot 2^{n-2} - n + 2.$$

$$= \left(3(-1)(-1)^{n-2} + 6(-1)^{n-2}\right) + \left(2\cdot 2^{n-2} + 2\cdot 2^{n-2}\right) - n + 2.$$

$$= (-1)^{n-2}\left(\overset{3}{\cancel{3(-1)+6}}\right) + 2^{n-2}\left(\overset{4=2^2}{\cancel{2+2}}\right) - n + 2.$$

$$= 3\cdot(-1)^{n-2} + 2^2 \times 2^{n-2} - n + 2.$$

$$= 3\cdot(-1)^{n-2} + 2^n - n + 2$$ #

6. (a).

6-(b)

let f as:

$$f: \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+, \quad f(m,n) = 2^m \cdot 3^n$$

check f is one-one:

if: $f(a,b) = f(m,n)$, then:

$$2^a \cdot 3^b = 2^m \cdot 3^n$$

$$\begin{array}{l} 2^a = 2^m \\ 3^b = 3^n \end{array} \Rightarrow \begin{array}{l} a = m, \\ b = n. \end{array}$$

$\Rightarrow$ f is a one-to-one.

$\therefore$ $|\mathbb{Z}^+ \times \mathbb{Z}^+| \leq |\mathbb{Z}^+|$.

$\Rightarrow$ a set is countable iff $|A| \leq |\mathbb{Z}^+|$

$\Rightarrow$ $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable.

7.

(a)

$$AB = \begin{bmatrix} 2 & -2 & -3 \\ 1 & 0 & 1 \\ 9 & -4 & 4 \end{bmatrix}.$$

(b)

$$A \vee B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A \wedge B = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

8.

$$f(x) = x^4 + 9x^3 + 4x + 7.$$

Def: If $f(x)$ is $O(g(x))$, $f(x) \leq Cg(x)$ $\forall x > k$.

when $x > 9$, $x^4 > 9x^3$, $x^4 > 4x$, $x^4 > 7$

$$f(x) = x^4 + 9x^3 + 4x + 7 < x^4 + x^4 + x^4 + x^4 = 4x^4$$

$\boxed{\text{let } C = 4,\ k = 9.}$

9.

$$f(x) = \frac{(x^2+1)}{(x+1)} \qquad g(x) = x.$$

by defn: if $f(x)$ is $O(g(x))$,

$\exists\, c, k$ s.t. $f(x) \leq C g(x)\ \forall x > k$

$$\begin{array}{r} x-1 \\ x+1 \overline{\smash{)}\, x^2+1} \\ \underline{x^2+x} \\ 1-x \\ \underline{-1-x} \\ 2. \end{array}$$

$\Rightarrow f(x) = (x-1) + \frac{2}{x+1}$

when $x > 0$: $x-1 < x$, $\frac{1}{x+1} < 1 \Rightarrow \frac{2}{x+1} < 2$.

$$f(x) = (x-1) + \frac{2}{x+1} < x+2 \underset{\downarrow}{<} x+x = 2x$$

when $x > 2$.

$\boxed{\text{let } C = 2,\ k = 2}$

10.

$f(x) = \dfrac{x^3+2x}{2x+1}$, $g(x) = x^2$.

by defn:

If $f(x)$ is $O(g(x))$, $\exists\, c, k$ s.t. $f(x) \le c \cdot g(x)$

$\forall x > k$.

$$
\begin{array}{r|l}
 & \frac{1}{2}x^2 - \frac{1}{4}x\ \frac{9}{8} \\
2x+1 & x^3 + 0x^2 + 2x + 0 \\
 & x^3 + \frac{1}{2}x^2 + 0 + 0 \\
\hline
 & -\frac{1}{2}x^2 + 2x + 0 \\
 & -\frac{1}{2}x^2 - \frac{1}{4}x + 0 \\
\hline
 & \frac{9}{4}x + 0 \\
 & \frac{9}{4}x + \frac{9}{8} \\
\hline
 & -\frac{9}{8}
\end{array}
$$

$\Rightarrow f(x) = \frac{1}{2}x^2 - \frac{1}{4}x + \frac{9}{8} - \dfrac{\frac{9}{8}}{2x+1}$

when $x > 2$, $x^2 > x > \frac{9}{8}$; when $x > 0$, $\frac{1}{4}x > 0$

$\dfrac{\frac{9}{8}}{2x+1} > 0$.

$f(x) = \frac{1}{2}x^2 \underbrace{- \frac{1}{4}x}_{>0} + \frac{9}{8} \underbrace{- \dfrac{\frac{9}{8}}{2x+1}}_{>0} < \frac{1}{2}x^2 + \underbrace{\frac{9}{8}}_{< x^2} < \frac{1}{2}x^2 + x^2 = \frac{3}{2}x^2$

$\boxed{\text{let } k = 2, \quad c = \frac{3}{2}}$